TD/TP 4 Supervised Learning 2A
2025-2026

Duration 3h
Use of ChatGPT / Bard / other for Part 2 encouraged.
This session consists of two parts:

- Part 1: TD to manipulate confusion matrices and ROC curves (2h)
- Part 2: TP to use these tools in the context of comparing discrimination models (1h)

Part 1: Confusion matrices, ROC curves

We consider the case of binary discrimination, where the output variable Y can take two values $Y = 1$ or $Y = 0$. We consider that the event of interest is $Y = 1$, and we have a training sample on which we trained a classifier. We also have a test sample.

Questions:

1. Assuming the classifier is fixed, recall how the confusion matrix is constructed.

2. For this same classifier, you are provided with the table below.

   2.1. What does the score variable correspond to? What are the "Individual" and "Observation" columns?

   2.2. What are the x-axis and y-axis of the ROC curve? How is it plotted?

   2.3. Add columns to the file to calculate the quantities you need for the ROC curve, then plot it.

   2.4. Add the curves corresponding to the perfect model and the random model.

3. We now focus on the ROC curve.

   3.1. What are the most interesting points on this curve?

   3.2. What does the area under the curve represent? What values can it take?

   3.3. How to compare two models using their ROC curves?

Individual	Observation Y	Score					
1	1	0,92825					
2	0	0,82762					
3	1	0,74119					
4	1	0,73182					
5	1	0,72418					
6	1	0,63537					
7	0	0,63155					
8	0	0,61763					
9	1	0,49700					
10	1	0,46686					
11	1	0,37062					
12	0	0,31244					
13	0	0,29258					
14	0	0,26446					
15	1	0,22945					
16	0	0,21983					
17	0	0,17818					
18	0	0,13490					
19	0	0,12012					
20	0	0,05337					
21	0	0,01124					
22	0	0,00639					
23	0	0,00241					
24	0	0,00102					
25	0	0,00099					
26	0	0,00040					
27	0	0,00011					

Part 2: Implementation in R

Recommended package for ROC curves: plotROC.

The aim of this part of the TP is spam detection in emails received by an individual: to build a filter that automatically distinguishes spam from relevant messages with a view to their non-delivery or even their destruction.

Spam includes "unsolicited" advertising for products, websites, tricks to earn money easily, invitations to forward a message to a certain number of people ("chains") and many other unwanted messages that pollute electronic mailboxes. It should be noted that the theme of spam received by an individual strongly depends on their network of contacts, their interests, their professional activity, etc... It is therefore possible to personalize a spam detector.

The database was compiled by a Hewlett-Packard engineer (George Forman, Palo Alto, USA) who collected a number of his relevant emails (non-spam) and others considered undesirable (spam) in June and July 1999. The spam filter we are going to build is therefore personalized because the learning is done on the messages he receives, the characteristics of the messages received reflect the messages received by this person.

The database contains 4601 messages, of which 1813 are spam (39.4%). 57 variables allow each message to be characterized, they correspond to the characters, words it contains and especially the frequency with which they appear:

- 48 continuous variables [0-100] are of the type "frequency_of_word_WORD": percentage of words in the email equal to WORD = 100*(number of times the word WORD appears in the e-mail / total number of words in the email). A WORD is here a sequence of alphanumeric characters bounded by non-alphanumeric characters or blanks. The relative frequencies of 48 particular words are calculated for each email
- 6 continuous variables [0-100] are of the type "frequency_of_character_CHARACTER": the same definition as above, with not words, but particular characters such as "!", for example.
- 1 positive continuous variable equal to the average length of uninterrupted sequences of capital letters in the message (V55)
- 1 positive continuous variable equal to the longest length of all uninterrupted sequences of capital letters (V56)
- 1 continuous variable equal to the sum of the lengths of all uninterrupted sequences of capital letters (=total number of capital letters in the message) (V57)
- **The target variable is the message status: is_spam=0 (non-spam) or is_spam =1 (spam)**

The data file is available in the "nutshell" library, it is the "spambase" table.

Implement the following methods: linear and quadratic discriminant analysis, CART tree, a CART tree with differentiated cost of 5 to 1, random forests, and k nearest neighbors. In particular, as in TP2, you will need to propose a procedure to optimize the hyper-parameters of the methods, if necessary.

Finally, compare using the ROC curve and the AUC the performance of these methods on this dataset and comment.